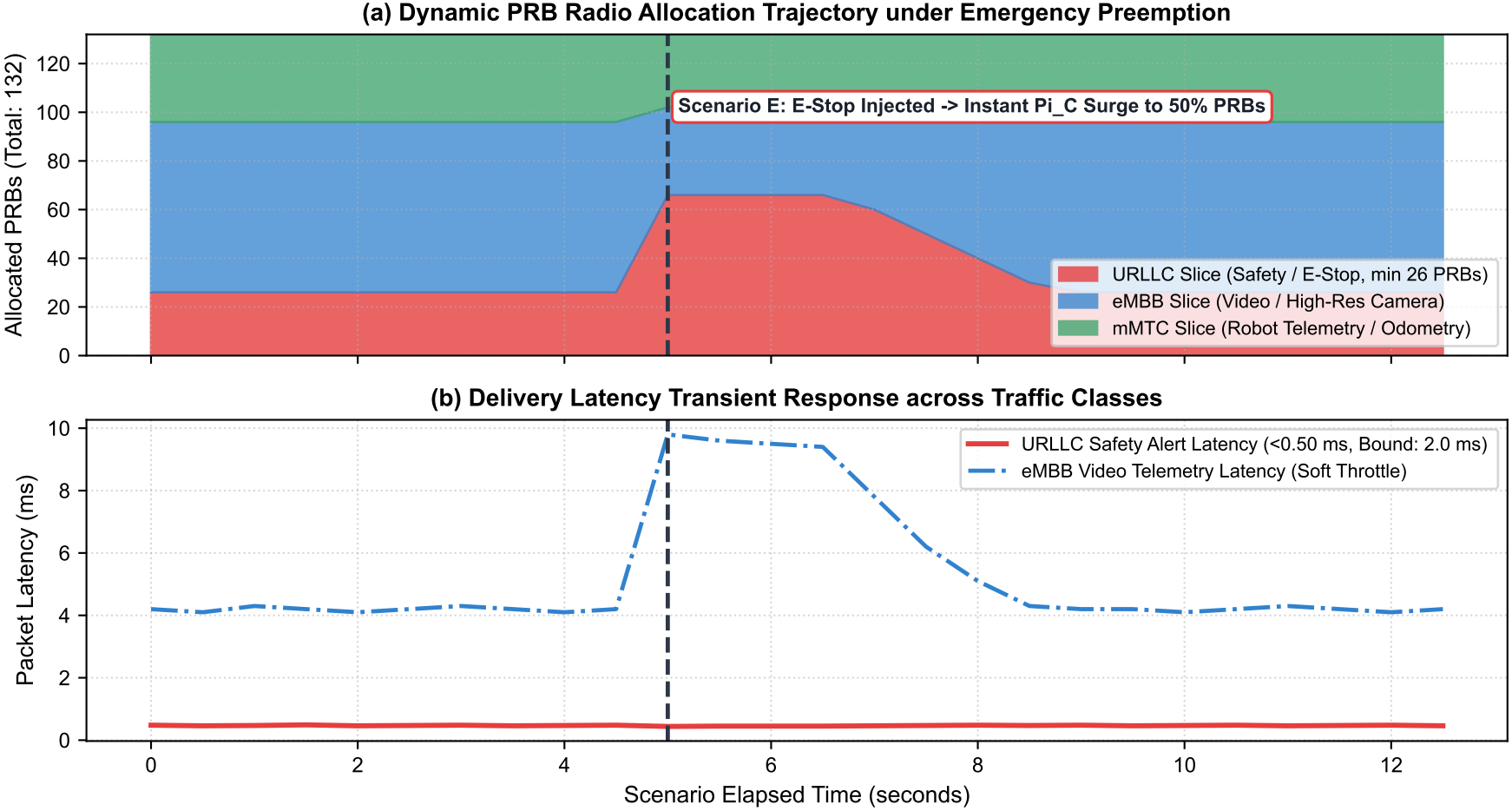
(a) Dynamic PRB Radio Allocation Trajectory under Emergency Preemption
Allocated PRBs (Total: 132)
0
20
40
60
80
100
120
Scenario E: E-Stop Injected -> Instant Pi_C Surge to 50% PRBs
URLLC Slice (Safety / E-Stop, min 26 PRBs)
eMBB Slice (Video / High-Res Camera)
mMTC Slice (Robot Telemetry / Odometry)
(b) Delivery Latency Transient Response across Traffic Classes
Packet Latency (ms)
0
2
4
6
8
10
URLLC Safety Alert Latency (<0.50 ms, Bound: 2.0 ms)
eMBB Video Telemetry Latency (Soft Throttle)
0
2
4
6
8
10
12
Scenario Elapsed Time (seconds)